Math 306 Exercise Set 7

Textbook exercises from Section 6.A: 4, 6, 7, 23, 24.
Textbook exercises from Section 6.B: 2, 3, 7, 8, 15.

1. Define trace A to be the sum of the diagonal entries in the square matrix A. Let $M_{m,n}(\mathbb{R})$ be the vector space of $m \times n$ matrices with entries in $\mathbb{R}$ and define $\langle A, B \rangle = \text{trace}(B^\top A)$.

1. Show that $\langle A, B \rangle$ is an inner product on $M_{m,n}(\mathbb{R})$.
2. Show that if $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ are written as $n \times 1$ matrices, then $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x} \cdot \mathbf{y}$ where $\cdot$ denotes the dot product.
3. Let $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. Find $\|A\|$ and the distance from A to the 3×3 identity matrix.
4. Let A be an $n \times n$ matrix. This exercise shows there is an invertible matrix B arbitrarily close to A. Let $\varepsilon > 0$ and take $\delta \in [0, \varepsilon/\sqrt{n})$ such that δ is not an eigenvalue of A. Show that $B = A - \delta I$ is an invertible matrix such that $\|A - B\| \leq \varepsilon$.

2. Use the triangle inequality to show that

$$\|x\| - \|y\| \leq \|x - y\|$$

for all $x, y \in V$. (Therefore if $f : V \to \mathbb{R}$ is defined by $f(x) = \|x\|$, then $f(x)$ and $f(y)$ are close whenever x and y are close; this is what it means for f to be a continuous function.)

3. Let $v_1, \ldots, v_n$ be a basis for an inner product space V and let A be the $n \times n$ matrix with i, j entry equal to $\langle v_i, v_j \rangle$. If $x = c_1 v_1 + \cdots + c_n v_n \in V$, let $\mathbf{x}$ be the $n \times 1$ matrix with i^{th} entry equal to c_i. Prove that

$$\langle x, y \rangle = \mathbf{x}^\top A \overline{\mathbf{y}}.$$